EXPERIMENT – 5

AIM: To create and manipulate SQL views for department salary summary and employee hierarchy, test the updatability of views, and implement a recursive CTE to display employee reporting chains.

SOURCE CODE:

USING EXPERIMENT – 3 DATABASE

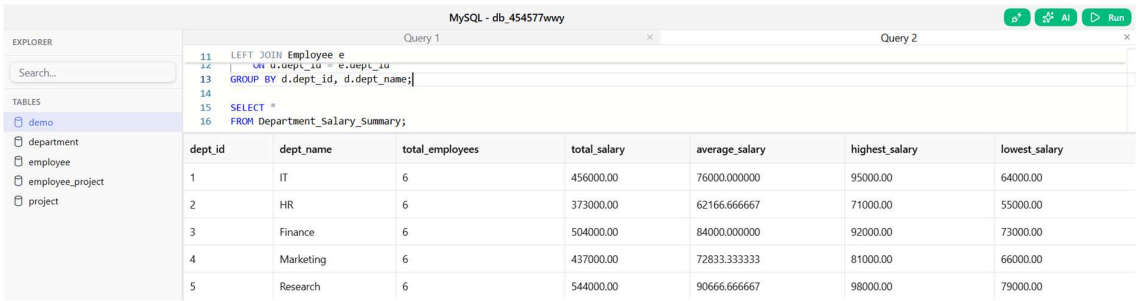
SQL QUERIES:

DEPARTMENT SALARY SUMMARY VIEW

```
CREATE VIEW Department_Salary_Summary AS
SELECT
    d.dept_id,
    d.dept_name,
    COUNT(e.emp_id) AS total_employees,
    SUM(e.salary) AS total_salary,
    AVG(e.salary) AS average_salary,
    MAX(e.salary) AS highest_salary,
    MIN(e.salary) AS lowest_salary
FROM Department d
LEFT JOIN Employee e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_id, d.dept_name;

SELECT *
FROM Department_Salary_Summary;
```

OUTPUT:

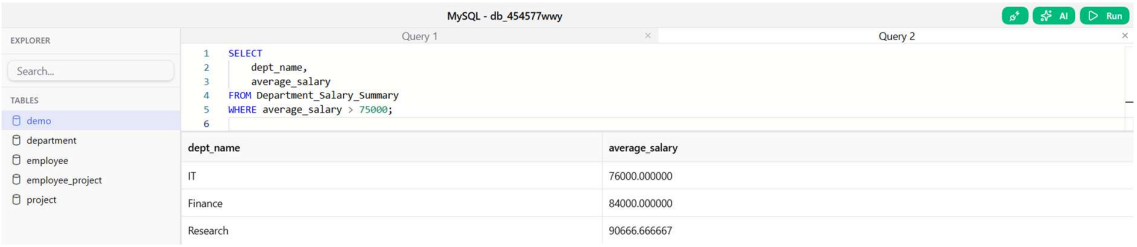


dept_id	dept_name	total_employees	total_salary	average_salary	highest_salary	lowest_salary
1	IT	6	456000.00	76000.000000	95000.00	64000.00
2	HR	6	373000.00	62166.666667	71000.00	55000.00
3	Finance	6	504000.00	84000.000000	92000.00	73000.00
4	Marketing	6	437000.00	72833.333333	81000.00	66000.00
5	Research	6	544000.00	90666.666667	98000.00	79000.00

QUERY THE VIEW

```
--Find departments with an average salary greater than ₹75,000
SELECT
    dept_name,
    average_salary
FROM Department_Salary_Summary
WHERE average_salary > 75000;
```

OUTPUT:



EMPLOYEE HIERARCHY

```
ALTER TABLE Employee
ADD manager_id INT NULL;

ALTER TABLE Employee
ADD CONSTRAINT fk_employee_manager
FOREIGN KEY (manager_id)
REFERENCES Employee(emp_id);
```

INSERT EMPLOYEE HIERARCHY

```
-- Update Manager IDs

UPDATE Employee SET manager_id = NULL WHERE emp_id IN
(105, 112, 114, 122, 125);

-- IT

UPDATE Employee
SET manager_id = 105
WHERE emp_id IN (103, 104);

UPDATE Employee
SET manager_id = 103
WHERE emp_id IN (101, 102);

UPDATE Employee
SET manager_id = 104
```

```
WHERE emp_id = 106;

-- HR

UPDATE Employee
SET manager_id = 105
WHERE emp_id IN (103, 104);

UPDATE Employee
SET manager_id = 103
WHERE emp_id IN (101, 102);

UPDATE Employee
SET manager_id = 104
WHERE emp_id = 106;

-- Finance

UPDATE Employee
SET manager_id = 112
WHERE emp_id IN (107, 108);

UPDATE Employee
SET manager_id = 107
WHERE emp_id = 109;

UPDATE Employee
SET manager_id = 108
WHERE emp_id IN (110, 111);

-- Marketing

UPDATE Employee
SET manager_id = 112
WHERE emp_id IN (107, 108);

UPDATE Employee
SET manager_id = 107
WHERE emp_id = 109;

UPDATE Employee
SET manager_id = 108
WHERE emp_id IN (110, 111);

-- Research

UPDATE Employee
SET manager_id = 125
WHERE emp_id IN (126, 127, 128);

UPDATE Employee
SET manager_id = 126
```

```
WHERE emp_id = 129;

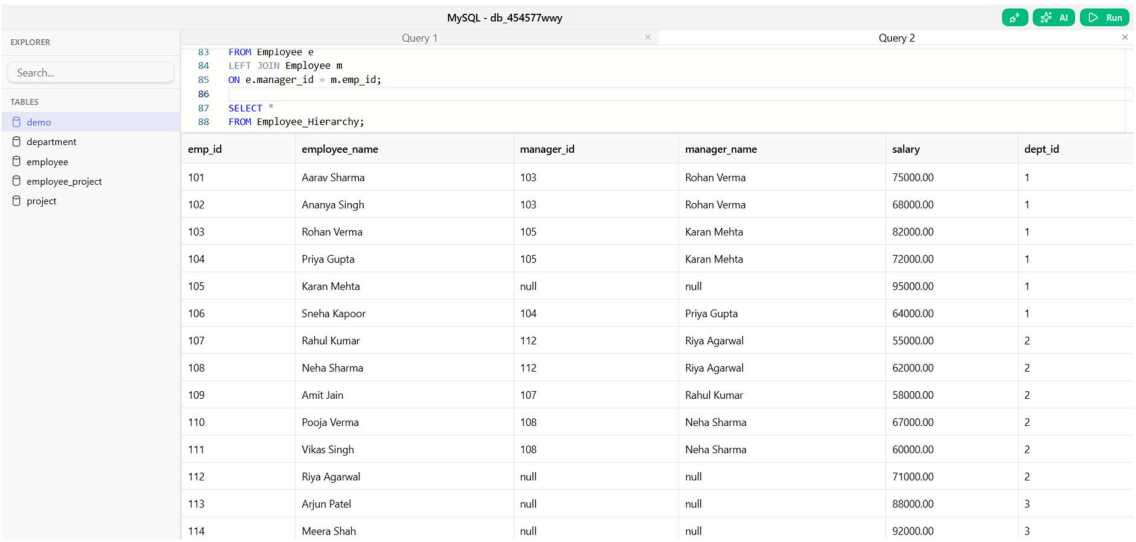
UPDATE Employee
SET manager_id = 128
WHERE emp_id = 130;
```

EMPLOYEE HIERARCHY VIEW

```
CREATE VIEW Employee_Hierarchy AS
SELECT
    e.emp_id,
    e.emp_name AS employee_name,
    e.manager_id,
    m.emp_name AS manager_name,
    e.salary,
    e.dept_id
FROM Employee e
LEFT JOIN Employee m
    ON e.manager_id = m.emp_id;

SELECT *
FROM Employee_Hierarchy;
```

OUTPUT:



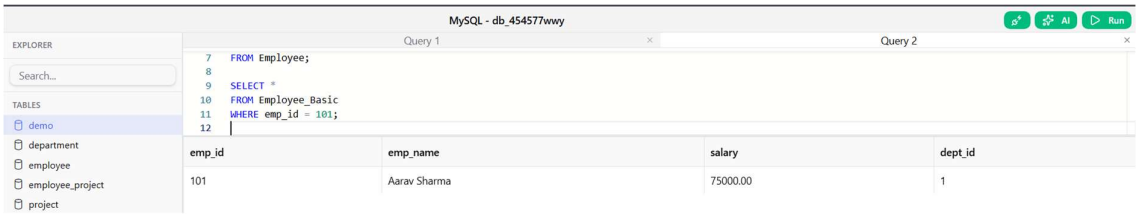
emp_id	employee_name	manager_id	manager_name	salary	dept_id
101	Aarav Sharma	103	Rohan Verma	75000.00	1
102	Ananya Singh	103	Rohan Verma	68000.00	1
103	Rohan Verma	105	Karan Mehta	82000.00	1
104	Priya Gupta	105	Karan Mehta	72000.00	1
105	Karan Mehta	null	null	95000.00	1
106	Sneha Kapoor	104	Priya Gupta	64000.00	1
107	Rahul Kumar	112	Riya Agarwal	55000.00	2
108	Neha Sharma	112	Riya Agarwal	62000.00	2
109	Amit Jain	107	Rahul Kumar	58000.00	2
110	Pooja Verma	108	Neha Sharma	67000.00	2
111	Vikas Singh	108	Neha Sharma	60000.00	2
112	Riya Agarwal	null	null	71000.00	2
113	Arjun Patel	null	null	88000.00	3
114	Meera Shah	null	null	92000.00	3

TEST UPDATABILITY OF A SINGLE VIEW

```
CREATE VIEW Employee_Basic AS
SELECT
    emp_id,
    emp_name,
    salary,
    dept_id
FROM Employee;

SELECT *
FROM Employee_Basic
WHERE emp_id = 101;
```

OUTPUT:

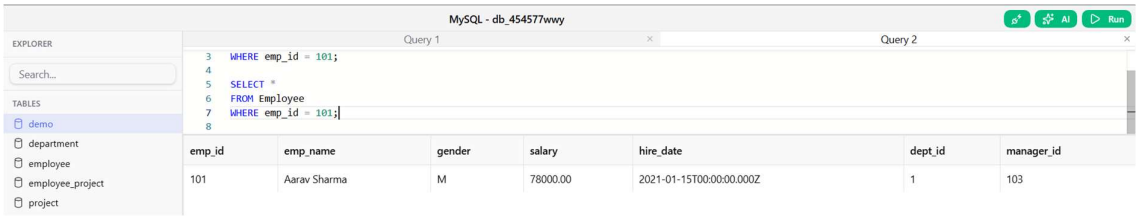


UPDATE THROUGH THE VIEW

```
UPDATE Employee_Basic
SET salary = 78000
WHERE emp_id = 101;

SELECT *
FROM Employee
WHERE emp_id = 101;
```

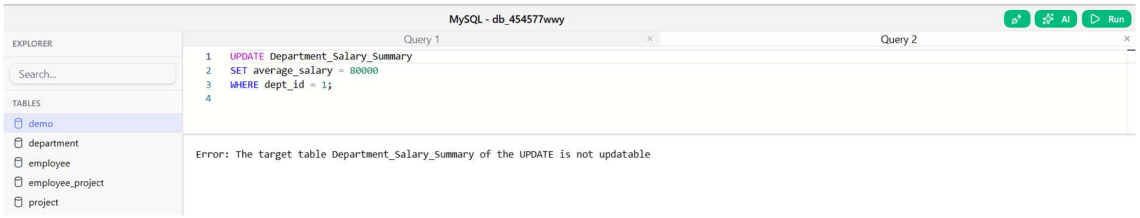
OUTPUT:



TEST NON-UPDATABILITY OF DEPARTMENT SALARY VIEW

```
UPDATE Department_Salary_Summary
SET average_salary = 80000
WHERE dept_id = 1;
```

OUTPUT:



RECURSIVE CTE

```
WITH RECURSIVE EmployeeChain AS
(
    -- Anchor member
    SELECT
        emp_id,
        emp_name,
        manager_id,
        0 AS level,
        CAST(emp_name AS CHAR(500)) AS reporting_chain
    FROM Employee
    WHERE manager_id IS NULL

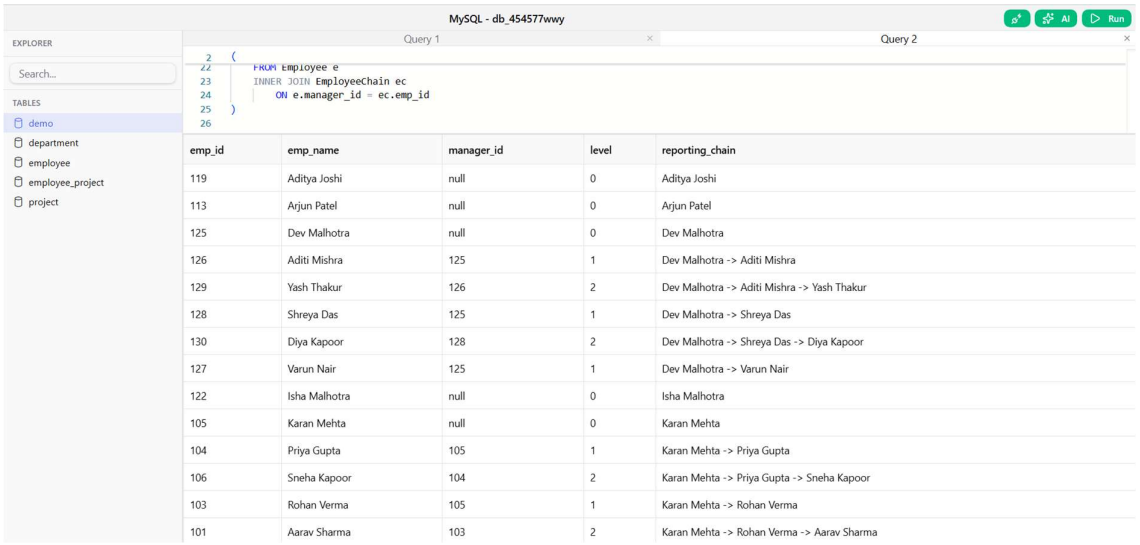
    UNION ALL

    -- Recursive member
    SELECT
        e.emp_id,
        e.emp_name,
        e.manager_id,
        ec.level + 1,
        CONCAT(ec.reporting_chain, ' -> ', e.emp_name)
    FROM Employee e
    INNER JOIN EmployeeChain ec
        ON e.manager_id = ec.emp_id
)

SELECT
    emp_id,
    emp_name,
    manager_id,
    level,
    reporting_chain
```

```
FROM EmployeeChain
ORDER BY reporting_chain;
```

OUTPUT:



emp_id	emp_name	manager_id	level	reporting_chain
119	Aditya Joshi	null	0	Aditya Joshi
113	Arjun Patel	null	0	Arjun Patel
125	Dev Malhotra	null	0	Dev Malhotra
126	Aditi Mishra	125	1	Dev Malhotra -> Aditi Mishra
129	Yash Thakur	126	2	Dev Malhotra -> Aditi Mishra -> Yash Thakur
128	Shreya Das	125	1	Dev Malhotra -> Shreya Das
130	Diya Kapoor	128	2	Dev Malhotra -> Shreya Das -> Diya Kapoor
127	Varun Nair	125	1	Dev Malhotra -> Varun Nair
122	Isha Malhotra	null	0	Isha Malhotra
105	Karan Mehta	null	0	Karan Mehta
104	Priya Gupta	105	1	Karan Mehta -> Priya Gupta
106	Sneha Kapoor	104	2	Karan Mehta -> Priya Gupta -> Sneha Kapoor
103	Rohan Verma	105	1	Karan Mehta -> Rohan Verma
101	Aarav Sharma	103	2	Karan Mehta -> Rohan Verma -> Aarav Sharma

RECURSIVE CTE FOR ONE EMPLOYEE

```
WITH RECURSIVE ReportingChain AS
(
    SELECT
        emp_id,
        emp_name,
        manager_id,
        0 AS level
    FROM Employee
    WHERE emp_id = 101

    UNION ALL

    SELECT
        e.emp_id,
        e.emp_name,
        e.manager_id,
        rc.level + 1
    FROM Employee e
    JOIN ReportingChain rc
        ON e.emp_id = rc.manager_id
)

SELECT
    emp_id,
    emp_name,
```

```
    manager_id,
    level
FROM ReportingChain;
```

OUTPUT:

